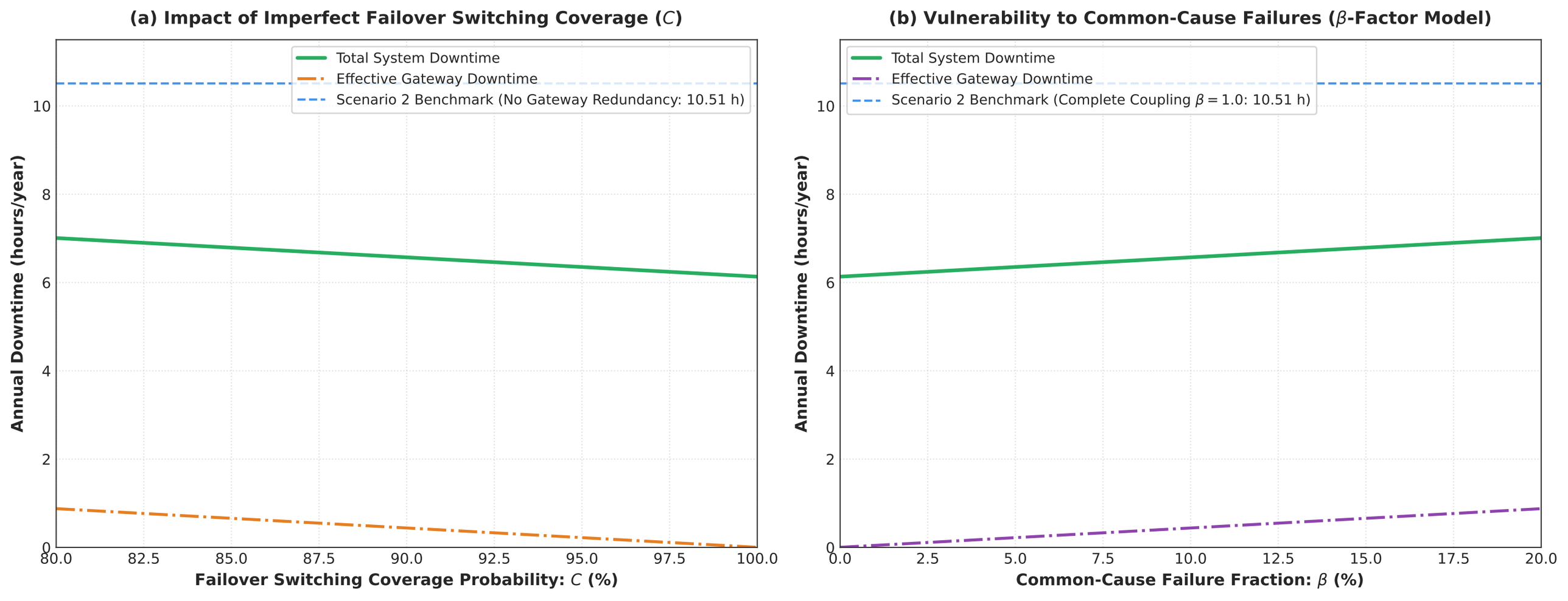
(a) Impact of Imperfect Failover Switching Coverage ($C$)
Total System Downtime
Effective Gateway Downtime
Scenario 2 Benchmark (No Gateway Redundancy: 10.51 h)
Annual Downtime (hours/year)
0
2
4
6
8
10
80.0
82.5
85.0
87.5
90.0
92.5
95.0
97.5
100.0
Failover Switching Coverage Probability: $C$ (%)
(b) Vulnerability to Common-Cause Failures ($\beta$-Factor Model)
Total System Downtime
Effective Gateway Downtime
Scenario 2 Benchmark (Complete Coupling $\beta = 1.0$: 10.51 h)
Annual Downtime (hours/year)
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Common-Cause Failure Fraction: $\beta$ (%)